The Supply of Shadow Banking, Bank Competition, and Real Investment in China

Abstract

Corporate shadow banking remains a challenge to Chinese financial development: firms that financialise engage less in real economic activity, and the opacity of the sector poses risks to financial stability. We address this challenge from the supply side by examining the role of bank competition in shaping firms' incentives to engage in shadow lending rather than invest in real economic activity, focusing on firm heterogeneity in particular. Using a model of firms' capital allocation under symmetric and asymmetric funding constraints that accounts for the moderating role of firms' capital structure, we show how bank competition changes firms' incentive structure toward real economic activity, thereby reducing their supply of shadow banking capital. Our model finds empirical validation in a panel of A-listed Chinese firms. Firm heterogeneity affects this relationship in important ways, which is particularly relevant for state-owned enterprises that play an important role in China's corporate funding environment.

JEL Classification: G21, G32, E44, G38, O16

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1. Introduction

Shadow banking has grown rapidly in China, reshaping how firms allocate capital between financial and productive uses. Much of the existing debate focuses on how shadow banking affects financial stability or the real economy (Gorton and Metrick, 2010; Pozsar et al., 2011; Acharya et al., 2013), yet a key, related question remains less explored: how does bank competition shape firms' engagement in shadow banking—through both supply- and demand-side channels—and what are the implications for real investment? This paper addresses the question by paying particular attention to the supply side of shadow banking—capital channelled by non-financial enterprises into entrusted loans, trust products, and other informal credit instruments—within China's distinctive institutional setting.

Bank competition matters because it alters firms' incentives to allocate capital between financial assets and real projects. In more competitive banking markets, greater credit access can reduce firms' reliance on shadow banking for funding, while tighter monitoring and lower arbitrage margins diminish firms' profitability on the supply side (Jayaratne and Strahan, 1996; Du et al., 2017; Liberti and Petersen, 2018; Allen et al., 2019). Conversely, in less competitive environments, constrained private firms may turn to shadow banking primarily as a source of credit to finance their activities, reflecting a demand-side channel

well documented in the literature (Jiang et al., 2023; Acharya et al., 2025). In addition, when credit constraints specifically limit real investment opportunities, some firms may divert available funds—whether retained earnings or partial borrowing—into shadow banking as an alternative use (Du et al., 2017). State-owned enterprises (SOEs), by contrast, often face few such constraints due to preferential credit access, and may use this advantage to actively supply shadow banking through re-lending and other financialised activities (Brandt and Li, 2003; Allen et al., 2005). How these forces play out in China's financial system—and whether competition can shift capital back toward productive use—remains an open question.

This question is particularly relevant in China's corporate lending environment, where a large share of shadow banking takes the form of entrusted loans—loans between firms arranged via a commercial bank that acts solely as a settlement and administrative agent without bearing credit risk. SOEs account for the majority of such lending[1] given their preferential access to bank credit, often justified by their special role in supporting the real economy. In practice, this creates a fragmented credit market in which some firms—mostly, but not exclusively, SOEs—obtain preferential bank funding and lend it on to more constrained firms, reinforcing asymmetric budget constraints that we address in both our theoretical and empirical analyses. These features further exhibit substantial regional variation. Differences in local financial development and in the transmission of collateral-based monetary policy imply that access to formal credit and the reliance on shadow intermediation vary across locations, reinforcing geographically heterogeneous credit segmentation (Liao et al., 2020; Yuan et al., 2024). This provides an institutional basis for examining bank competition at the regional level as a determinant of firms' shadow-banking activity, which we incorporate into our empirical strategy.

Policy so far had mixed effects on China's shadow banking sector: While a series of regulatory reforms between 2017 and 2018 sought to curb regulatory arbitrage in China's shadow banking sector, tightening oversight of entrusted loans, wealth management products, and trust lending,[2] China's monetary policy framework increasingly relies on collateral-based instruments such as the medium-term lending facility (MLF), which links liquidity provision to eligible collateral and thereby favours collateral-rich borrowers, setting incentives for sustained shadow banking (Le et al., 2022; Yuan et al., 2025). In this context, shadow banking continues to account for a significant share of corporate financing, with SOEs remaining central actors (Li et al., 2014). This persistence further suggests that institutional and market structural features—such as the dominance of a state-owned banking sector and the privileged position of SOEs—shape shadow banking outcomes in ways that targeted regulation of shadow banking alone may not fully address. In this paper, we argue that, against this context, corporate shadow banking is a feature rather than a bug: it reflects incentives embedded in the current market structure. Addressing it requires reforms that alter

[1] Allen et al. (2019) estimate a 64% share in transaction-level data between 2004 and 2013, far above comparable categories in other economies, including emerging markets. While this share has likely declined since 2013, SOEs remain dominant providers of entrusted loans.

[2] See: Sun (2019) as well as State Council (2018a) and State Council (2018b). See also Pistor (2019) for a broader discussion of China's financial governance framework.

these incentivesparticularly by increasing bank competition—to shift capital from shadow banking toward real economic activity.

To show this, we develop a stylised model in which bank competition can rein in opaque shadow lending practices and incentivise real economic activity by weakening credit constraints, particularly for private enterprises. The effect of bank competition is conditional on firms' internal capital structure: real and financial investment ratios moderate the shadow banking response through distinct mechanisms. High real investment intensity strengthens the negative effect of bank competition on shadow banking—acting as a pull factor toward productive investment and away from financial arbitrage—whereas high financial investment dampens this effect, acting as a push factor into shadow banking: firms with stronger financial orientation[3] operate further along the concave return curve, making them less sensitive to shrinking arbitrage margins and more reluctant to reallocate due to higher opportunity costs of shifting toward real investment. Considering both moderators separately is important because firms can exhibit high or low levels of each, making them economically distinct channels. This framework allows us to capture nuanced heterogeneity in how bank competition affects shadow banking.

Building on this theoretical framework, we empirically examine the predicted channels using data on Chinese A-share listed firms from 2011 to 2020. In particular, we test how the effect of bank competition on shadow banking depends on firms' capital structure, incorporating interaction terms between bank concentration, an inverse measure for competition,[4] and both financial and real investment ratios. The estimated moderating effects align with our model's predictions. We then conduct an extensive heterogeneity analysis—focusing especially on the role of SOEs—which highlights the importance of asymmetric budget constraints in shaping shadow banking activity. The results are robust to a range of exercises, including an instrumental-variable approach, a panel VAR specification to address potential endogeneity, and alternative estimators and fixed-effects structures. These findings have broader implications: if SOEs are intended to support the real economy, the persistence of their shadow banking activities calls for closer scrutiny. Our analysis thus documents key institutional determinants of corporate shadow lending across ownership types, offering evidence relevant for policy debates on aligning corporate lending behaviour with real-sector objectives.

Our paper contributes to the literature on bank competition, corporate financialisation, and shadow banking. Prior studies have examined how bank competition influences credit allocation and risk-taking (Jiang et al., 2020; Wang et al., 2023), as well as the demand side of shadow banking, where constrained firms rely on informal credit channels such as entrusted loans or non-bank intermediaries (Jiang et al., 2023; Acharya et al., 2025). A smaller but growing body of work has begun to explore supply-side channels, in which

[3]Financial orientation refers to the extent to which firms allocate resources toward financial assets relative to productive investment. Empirically, this is proxied by the share of financial assets in total assets, capturing firms' overall exposure to financial activities and their position along the financial intermediation margin.

[4]Bank concentration (e.g. the Herfindahl-Hirschman Index, HHI) is a common inverse proxy for bank competition, based on the premise that more concentrated markets exhibit less competitive pressure (Berger et al., 2004; Beck et al., 2006; Allen et al., 2019), which is satisfied in our case.

non-financial enterprises act as shadow credit providers by reallocating internal capital to financial uses (Allen et al., 2019; Zhuang et al., 2026). However, while several empirical studies document these patterns, there has been little theoretical formalisation of the mechanisms linking bank competition, firms' capital structure, and their incentives to engage in shadow banking. Existing models of bank competition and investment allocation (e.g., Rajan and Zingales, 1996; Holmstrom and Tirole, 1997; Boyd and De Nicolo, 2005) provide useful foundations but have not been extended to capture the specific institutional frictions and dual-track credit system that characterise China's corporate lending environment. This gap is particularly relevant given that the theoretical literature on bank competition offers competing predictions: one view holds that greater competition may increase risk-taking by reducing bank margins, weakening monitoring, and exacerbating moral hazard (Berger et al., 2020; Naili and Lahrichi, 2022), while another suggests that competition improves allocative efficiency, relaxes financing constraints, and enhances borrower screening (Jayaratne and Strahan, 1996; Liberti and Petersen, 2018), thereby reducing incentives for firms to divert capital into financial assets and instead encouraging productive investment (Demir, 2009; Cornaggia et al., 2015). Our contribution helps to reconcile these perspectives by highlighting how the effects of bank competition depend on firms' internal capital structures.

Against this backdrop, this paper makes three contributions. First, we provide a novel supply-side perspective on shadow banking: whereas most prior studies emphasise how credit constraints drive shadow borrowing, we show—both theoretically and empirically—that bank competition curtails firms' role as shadow credit suppliers by encouraging reallocation toward the real economy. Second, we develop a stylised model that formalises how internal capital structure—specifically, the allocation of resources to financial versus productive uses—moderates the effect of bank competition on shadow banking. This framework helps reconcile competing views in the literature by showing that the same competitive shock can trigger different responses depending on a firm's financial orientation. Third, we contribute to the literature on banking deregulation and sustainable growth by demonstrating that stronger competition can reduce the financialisation of non-financial firms, particularly in contexts like China where credit access is institutionally skewed and firms' investment incentives vary systematically by ownership.

The remainder of the paper proceeds as follows. Section 2 develops the theoretical framework and testable hypotheses. Section 3 describes the data, sample, and variable construction. Section 4 presents baseline results and robustness checks. Section 5 explores heterogeneity in the effects of bank competition across regulatory environments, ownership, and governance structures. Section 7 concludes.

2. Theoretical framework and hypotheses

We employ a stylized model inspired by Rajan and Zingales (1996), Holmstrom and Tirole (1997), and Boyd and De Nicolo (2005) to examine how bank competition affects the trade-off between real and financial investment by non-financial firms. The model offers a conceptual framework for understanding financialisation as a supply-side channel of shadow banking. This framework helps to quantify the allocative inefficiencies that arise when

capital intended for productive investment is diverted into shadow financial channels such as entrusted loans or structured products.

We distinguish between two roles firms may play in the shadow banking system: on the demand side, firms access shadow finance when formal bank credit is limited; on the supply side, firms channel surplus capital into shadow banking by acting as lenders. While both mechanisms are relevant, our analysis centers on the supply side—how firms allocate internal resources to shadow banking in response to changes in the banking environment. For this, we derive and present main hypotheses from our model below with more detailed proofs given in Appendix A.

2.1. Capital allocation with bank competition

A representative firm receives capital R, which it allocates between real investment I_r and financial investment I_f, subject to the budget constraint:

$$R = I_r + I_f.^{5}$$

The firm maximizes utility from investment:

$$\max_{I_r, I_f} \Pi = \phi(I_r) + \theta(I_f),$$

where $\phi(\cdot)$ and $\theta(\cdot)$ are strictly concave returns to real and financial investment, respectively. That is, $\phi' > 0$, $\phi'' < 0$ and $\theta' > 0$, $\theta'' < 0$.

Firms face a borrowing constraint on real investment,

$$I_r \leq \lambda(\kappa), \qquad \lambda'(\kappa) > 0,$$

where $\lambda(\kappa)$ denotes the amount of external finance available to the firm. Greater competition relaxes this constraint by expanding credit supply, reducing collateral requirements, and improving loan terms, thereby easing financing frictions for borrowers (Jayaratne and Strahan, 1996; Liberti and Petersen, 2018).

To make the financial channel explicit, we model returns to financial investment as

$$\theta(I_f, \kappa) = m(\kappa) I_f - \frac{\gamma}{2} I_f^2, \qquad \gamma > 0,$$

where $m(\kappa)$ denotes the intermediation margin associated with shadow-banking activity.

[5] Firms' assumed capital structure is a simplification that we employ to isolate the specific allocation margin through which bank competition affects corporate shadow-banking activity and real investment. The aim is not to provide a full representation of firms' balance sheets, but to capture allocations insofar as they affect firms' participation in shadow intermediation. This is consistent with empirical evidence for China, which emphasises the role of entrusted lending as a key form of firm-level financial intermediation linked to the banking system (Allen et al., 2019; Acharya et al., 2025). By contrast, alternative uses of funds, such as cash holdings or passive financial assets, are typically associated with liquidity management and precautionary motives rather than participation in the shadow banking intermediation channel.

We define

$$m(\kappa) = r_\ell(\kappa) - r_b(\kappa) - c(\kappa),$$

where $r_\ell(\kappa)$ is the return obtainable through shadow intermediation, $r_b(\kappa)$ is the effective funding cost, and $c(\kappa)$ captures monitoring and enforcement costs.

Bank competition affects this margin through two mechanisms. First, competitive pressure compresses the wedge between funding costs and re-lending returns by reducing lending margins and weakening segmentation in credit allocation. Second, increased competition strengthens scrutiny over credit transactions, raising the effective costs of shadow intermediation. Together, these effects imply

$$m'(\kappa) < 0,$$

so that the marginal return to financial investment declines as bank competition intensifies.

The model therefore focuses on financial activities whose profitability depends on this intermediation margin. It abstracts from passive portfolio assets, whose returns may vary with broader financial conditions but do not arise from shadow intermediation and therefore do not generate the competition-sensitive rents central to the mechanism studied here.

In the model, these mechanisms imply that firms reduce capital allocated to financial uses as bank competition intensifies,

$$\frac{dI_f^*}{d\kappa} < 0.$$

This yields our first testable implication:

H1. Shadow banking scale. Bank concentration increases shadow banking activity by non-financial firms,

$$\frac{d\, I_{f_{i,t}}^{SB}}{d\, HHI_{c,t}} > 0.$$

That is, for a given firm i in year t, an increase in local bank concentration, $HHI_{c,t}$, is associated with a rise in the firm's shadow banking activity, denoted by $I_{f_{i,t}}^{SB}$. The measure of shadow banking employed captures the firm-level supply of shadow credit through entrusted loans, intercorporate lending, and related instruments.[6] Bank concentration, measured via the Herfindahl-Hirschman Index (HHI), serves as an inverse proxy for bank competition: higher HHI reflects greater market power and reduced competition among banks. This prediction serves as a baseline for more nuanced hypotheses, which we explore further below. While it aligns with Jiang et al. (2023), our analysis extends beyond shadow bank borrowing to also capture supply-side dynamics at the firm level. We test this empirically by regressing bank concentration on firm-level shadow banking activity, estimating β_1 in equation (2).

[6] In line with the theoretical model, we interpret capital allocated to financial uses as shadow banking activity, which implicitly assumes that non-financial firms do not allocate substantial resources to other financial assets such as equities or bonds. This simplification is necessary to avoid introducing a full portfolio allocation structure into the model, and it is empirically plausible in the Chinese context, where corporate investment in non-bank financial markets remains limited and bank-linked instruments dominate firm-level financial activity.

2.2. Credit constraints and investment behavior

We next distinguish between cases where the credit constraint is binding and non-binding. In the non-binding case, firms equate marginal returns across real and financial investment, so that in equilibrium real investment, I_r^*, becomes:

$$\phi'(I_r^*) = \theta'(R - I_r^*).$$

Here, bank competition affects shadow banking indirectly by altering the relative returns, but does not influence the firm's capital availability: increased competition influences allocation only by lowering the relative profitability of financial investment, prompting a shift toward productive uses without changing total capital availability.

In the binding case, where $I_r^* > \lambda(\kappa)$, the constraint restricts real investment:

$$I_r^* = \lambda(\kappa), \quad I_f^* = R - \lambda(\kappa).$$

Firms cannot fully fund productive investments, and residual capital is diverted to shadow banking. In the presence of binding credit constraints, even profitable real projects may remain unfunded due to limited external financing, while internal funds—whether retained earnings or partially unused loan proceeds—are instead allocated to higher-yielding financial assets such as entrusted loans or trust products. Du et al. (2017) document such behaviour in China and Russia, where constrained firms re-lend surplus funds in the shadow banking market rather than expand physical capital investment, illustrating how limited credit availability can redirect resources into financialised activities. In this setting, bank competition reduces shadow banking by relaxing credit constraints and allowing greater productive investment.

Crucially, how strongly firms respond to changes in bank competition depends on their capital structure. We denote a firms' financial orientation by $\bar{I}_f$, which captures the degree to which it tends to allocate capital toward financial uses, and its real investment tendency by $\bar{I}_r$, defined analogously for productive investment. These parameters are conceptually distinct—firms can exhibit high or low values for both, depending on strategic orientation and institutional context—and they shape the channels through which bank competition influences shadow banking. A high $\bar{I}_f$ reflects an internal preference for financialised operations and a lower opportunity cost of maintaining them. Conversely, a high $\bar{I}_r$ reflects a stronger focus on productive capital and a higher opportunity cost of idle or misallocated capital.

This heterogeneity gives rise to two moderating mechanisms: financial orientation acts as a push factor into shadow banking, while real investment focus serves as a pull factor out of it. To formalize this, we examine how the marginal effect of competition on shadow banking, $\frac{dI_f^*}{d\kappa}$, varies with $\bar{I}_f$ and $\bar{I}_r$. Differentiating yields:

$$\frac{d^2 I_f^*}{d\kappa \, d\bar{I}_f} < 0, \quad \frac{d^2 I_f^*}{d\kappa \, d\bar{I}_r} > 0.$$

The first expression captures the financial push effect: firms with higher $\bar{I}_f$ are more deeply engaged in financial activities and therefore less responsive to changes in bank competition. Because they are already strongly oriented toward financial uses, marginal reductions in arbitrage returns—caused by intensified competition—result in smaller shifts in allocation. This implies a dampened reallocation response as financial orientation increases. The second expression reflects a real-side pull effect: firms with higher $\bar{I}_r$ face greater marginal benefits from productive investment and are thus more likely to shift capital out of shadow banking when financing frictions ease. In both cases, the marginal effect of competition on shadow banking depends on firms' internal capital allocation preferences, forming the theoretical basis for Hypotheses H2a and H2b.

H2. Supply-side channels.

H2a *Financial Orientation.* Conditional on a positive baseline relationship between bank concentration and shadow banking ($\beta_1 > 0$), higher financial orientation amplifies the effect of bank concentration on shadow banking:

$$\beta_1 > 0, \qquad \frac{d^2 I_f^{SB}}{dHHI d\tilde{I}_f} > 0.$$

H2b *Real Orientation.* Conditional on a negative baseline relationship between bank concentration and shadow banking ($\beta_1 < 0$), higher real investment intensity dampens the negative effect of bank concentration on shadow banking:

$$\beta_1 < 0, \qquad \frac{d^2 I_f^{SB}}{dHHI d\tilde{I}_r} > 0.$$

Empirically, we test H2a and H2b by interacting bank concentration $HHI_{c,t}$ with both the financial investment ratio $\tilde{I}_f$ and the real investment ratio $\tilde{I}_r$. These interaction terms capture how the marginal effect of bank concentration on shadow banking varies with firms' capital structure. Interpreting the interaction coefficient β_3 requires conditioning on the baseline effect of HHI, denoted β_1. Specifically, the total marginal effect of concentration becomes:

$$\frac{dI_f^{SB}}{dHHI} = \beta_1 + \beta_3 \cdot Z.$$

When $Z = \tilde{I}_f$, a positive interaction coefficient β_3 suggests that more financialised firms exhibit a smaller marginal response to bank concentration—consistent with H2a—*only if* the baseline effect β_1 is positive (i.e., greater concentration leads to more shadow banking). Similarly, when $Z = \tilde{I}_r$, a positive β_3 indicates a stronger marginal effect for capital-intensive firms *if* β_1 is positive—supporting H2b. Thus, the direction and strength of the interaction must be interpreted jointly with the level effect. In both cases, internal capital structure moderates the influence of banking market structure on shadow banking, aligning with the theoretical predictions of financial push and real pull channels.

2.3. Constraint asymmetry and institutional distortion

China's institutional setting gives rise to pronounced asymmetries in firms' access to credit. State-owned enterprises (SOEs) benefit from preferential lending policies and implicit guarantees (e.g., Brandt and Li, 2003; Allen et al., 2005), weakening the disciplining effect of bank competition. In our model, this is reflected by looser borrowing constraints for SOEs:

$$\lambda_{\text{SOE}}(\kappa) > \lambda_{\text{Private}}(\kappa) \quad \forall \kappa.$$

Because private firms face tighter credit limits, they respond more strongly to marginal increases in competition by reallocating capital from shadow banking into real investment. SOEs, by contrast, are less constrained and may divert surplus liquidity into financial assets, such as entrusted loans, sustaining shadow banking activity. This results in the following first-order comparative statics:

$$\frac{dI_f^*}{d\kappa}\Big|_{\text{Private}} < \frac{dI_f^*}{d\kappa}\Big|_{\text{SOE}}, \quad \frac{dI_r^*}{d\kappa}\Big|_{\text{Private}} > \frac{dI_r^*}{d\kappa}\Big|_{\text{SOE}}.$$

As discussed in the previous section, these first-order effects are shaped by the presence of push and pull mechanisms—with financialised firms more inclined to supply shadow banking capital (push) and capital-intensive firms more likely to reallocate into productive use when constraints ease (pull). What distinguishes SOEs is that their marginal responsiveness to competition becomes more sensitive to internal capital allocation.

For firms with high financial investment, the key mechanism is again the push effect, where bank competition erodes arbitrage margins, and the incentive to maintain high levels of financial investment weakens. Following the logic from above, this push effect is subject to diminishing returns and highly financialised firms are less sensitive to changes in bank competition. This channel is particularly relevant for SOEs, which are disproportionately exposed to financial activities. Since they operate further along the concave financial return function $\theta(I_f)$, small changes in competition lead to smaller marginal shifts in shadow banking, implying weaker responsiveness when financialisation is high. In contrast, firms with high real investment exhibit a pull effect. As competition increases, financing constraints ease, and the productive use of capital becomes relatively more attractive. For capital-intensive firms, the opportunity cost of holding idle or misallocated capital in shadow banking is higher, especially as they have more real opportunities to pursue. Moreover, the marginal benefit of real investment effectively increases as competition opens access to productive financing channels. For SOEs that are already real-intensive, these effects are amplified: they have the internal liquidity and institutional support to reallocate efficiently when credit frictions diminish. As a result, the marginal effect of competition on shadow banking is stronger among capital-intensive SOEs.

Together, these mechanisms imply that while SOEs may react less overall to competition due to softer constraints but the sensitivity of their response depends more strongly on internal capital allocation. In other words, the disciplining effect of competition becomes more pronounced for SOEs that are either highly invested in the real economy (due to increasing marginal benefits and lower opportunity costs) or highly financialised (due to greater baseline

exposure to arbitrage incentives). This leads to the second-order comparative statics:

$$\frac{d^2 I_f^*}{d\kappa\, d\bar{I}_f}\Big|_{\text{SOE}} < \frac{d^2 I_f^*}{d\kappa\, d\bar{I}_f}\Big|_{\text{Private}}, \quad \frac{d^2 I_f^*}{d\kappa\, d\bar{I}_r}\Big|_{\text{SOE}} > \frac{d^2 I_f^*}{d\kappa\, d\bar{I}_r}\Big|_{\text{Private}}.$$

These second-order effects form the basis for Hypothesis H3: the marginal effect of bank concentration on shadow banking, conditional on financial or real investment, differs systematically by ownership type.

H3. Firm ownership. The effect of bank concentration on shadow banking, conditional on financial or real investment, is more sensitive in state-owned enterprises than in private firms:

$$\frac{d^2 I_f^{SB}}{d\,HHI\, d\,\tilde{I}_f}\Big|_{\text{SOE}} < \frac{d^2 I_f^{SB}}{d\,HHI\, d\,\tilde{I}_f}\Big|_{\text{Private}}, \quad \frac{d^2 I_f^{SB}}{d\,HHI\, d\,\tilde{I}_r}\Big|_{\text{SOE}} > \frac{d^2 I_f^{SB}}{d\,HHI\, d\,\tilde{I}_r}\Big|_{\text{Private}}.$$

We test this hypothesis by estimating how the relationship between bank concentration and shadow banking varies across firm ownership types, focusing on how this relationship is moderated by firms' investment behaviour. Specifically, we compare the estimated effects of bank concentration on shadow banking for firms with different levels of financial and real investment, separately for SOEs and private firms. These effects are captured through interaction terms between bank concentration and financial investment, and between bank concentration and real investment, estimated within each ownership group. The interaction terms are stronger and more significant for SOEs in the real investment dimension, but weaker or insignificant in the financial dimension. This indicates that capital-intensive state-owned firms respond more strongly to changes in banking market structure by reallocating toward productive investment, while highly financialised SOEs exhibit weaker marginal responses due to diminishing returns. These results suggest that institutional incentives and credit access asymmetries amplify SOEs' responsiveness to pull-based reallocation, but dampen push-channel responsiveness when financial engagement is already high.

3. Data and variables

We employ annual firm-level data of all publicly listed companies in China's A-share market from 2011 to 2020,[7] using the CSMAR database, supplemented by data sourced from the Wind database. Data on bank competition are compiled from the financial institution license records of the China Banking and Insurance Regulatory Commission, following Degryse and Ongena (2007) and Du and Cheng (2022).

Given the focus of our paper on non-financial firms and following common approaches in the literature we exclude firms, (1) that operate in the insurance and financial industries; (2)

[7] We exclude data prior to 2011 and after 2020 dynamics relating to the aftermath of the financial crisis and the COVID pandemic may affect the stability of our estimates in a way that does not reflect the mechanism we aim at estimating in this paper.

that have missing values for key variables; (3) that were classified as ST (Special Treatment) or *ST (Special Treatment with a warning of delisting) or PT (delisted) during the sample period. To mitigate the influence of extreme values, we further winsorized all continuous variables at the 1st and 99th percentiles. In the end, this process resulted in an unbalanced panel with a total of 25,071 firm-year observations and 3738 firms for the main variables.

3.1. Variables

3.1.1. Shadow banking

In line with Han et al. (2023), we assume that enterprises engage in shadow banking activities directly through entrusted loans, entrusted wealth management, and private borrowing, i.e., off-balance-sheet financing channels that operate outside traditional bank lending. These items represent claims arising from firm-level credit provision and therefore capture the supply of shadow banking activity rather than general financial asset holdings. Entrusted loans refer to firm-to-firm lending intermediated by commercial banks, while entrusted wealth management involves firms investing surplus funds in bank-issued or third-party wealth management products. Private borrowing typically includes informal lending arrangements that bypass formal credit institutions. These channels allow firms to extend or obtain credit with greater flexibility but less regulatory oversight. We use the natural logarithm of the total scale of all these activities ($\ln I_f^{SB}$) to measure the level of shadow banking in non-financial firms.

Firm-level shadow banking activity is not confined to a single balance-sheet item but spans several closely related financial claims, including entrusted loans, intercorporate lending, and related receivables. These items reflect different manifestations of the same underlying activity—firm-level credit intermediation—and therefore jointly capture the supply of shadow credit. We consequently construct a composite measure to reflect the multidimensional nature of shadow banking in the Chinese context.

Given that entrusted loans constitute one of the largest components of China's shadow banking system (Allen et al., 2019), we use the proportion of entrusted loans from non-financial enterprises to total assets (I_f^{EL}) as a proxy variable for shadow banking in the robustness test section. The amounts of entrusted loans and financial product purchases (e.g., bank wealth management) are obtained from the "other current assets" category in the CSMAR database. The amount of entrusted wealth management is obtained from the CSMAR's external investment database. Due to its high level of secrecy, following the method of Han et al. (2023), the "other receivables" category in the enterprises' financial statements is used as a proxy variable for private borrowing, with data sourced from the financial statements of listed companies.

3.1.2. Bank competition

The core independent variable in this paper is bank competition, which is measured as bank concentration using the Herfindahl-Hirschman Index. The index captures city-level bank concentration based on each bank's market share in a city.[8] Here more concentration

[8]Firm location is defined based on the registered headquarters reported in the CSMAR database, which we use to match firms to local banking market conditions. While some listed firms operate across multiple

implies that fewer banks dominate, which reduces competitive pressure on lenders. We employ this measure as it captures the local concentration of lending power at the city level—the relevant scope for firms' financing decisions—and reflects both the dominance and fragmentation of the banking sector in a way that aligns with our hypotheses about firm behaviour under competitive pressure. Specifically, HHI increases sharply when a small number of banks control most of the market (indicating dominance), and decreases when market shares are more evenly distributed across many banks (indicating fragmentation). This mirrors our theoretical mechanism: when bank competition intensifies (i.e., HHI falls), firms face tighter credit scrutiny and reduced arbitrage opportunities, especially if they are highly financialised or not capital-intensive—thereby dampening their shadow banking activity. Using city-level data in measures for bank competition is further important, given the spacial variance in Chinese data that is particularly pronounced on a city level, whilst province level data cannot capture the same variation. Following the approach of Jiang et al. (2019) and Huang et al. (2021), we then consider data for five categories of commercial banks in the construction of our bank competition measures: city commercial banks, joint-stock commercial banks, state-owned banks, private banks, and foreign banks. The Herfindahl-Hirschman Index then reflects the industry concentration of the regional banking sector and is inversely related to bank competition. In other words, a higher HHI indicates lower levels of banking competition:

$$HHI = \sum_{n=1}^{n} \left(\frac{Branch_n}{Total_{Branch}}\right)^2, \tag{1}$$

where $0 \leq HHI \leq 1$, $Branch_n$ represents the total number of branch offices for the n-th bank in the region, and $Total_{Branch}$ represents the total number of branch offices for all banks in the region.

We further employ a province-level measure for bank competition, HHI^p, in addition to our main city-level measures as this captures competition among banks that may not be confined to competition within a single city: There may be spillover effects, for example in regions with relatively less developed financial infrastructure, businesses might need to seek banking services from other cities. If competition is too intense locally, commercial banks may also actively expand their operations to other cities. Finally, regulatory restrictions imposed by the China Banking and Insurance Regulatory Commission require commercial banks that seek to open secondary branches in a city to have their application reviewed and approved by the provincial-level regulatory authority under which the proposed branch falls. With this the regulatory authority in a province plays a crucial role in determining the level of competition among banks within that province.

3.1.3. Financial and real economic activity

In line with the above, bank competition may influence firms' participation in both shadow banking and the real economy. Du et al. (2017) document re-lending practices

locations, banking relationships and credit conditions are typically tied to the place of registration. To the extent that multi-city operations introduce measurement noise, this would attenuate the estimated effects rather than generate spurious relationships.

by firms—companies that redirect surplus or borrowed funds into informal lending channels—particularly in China and Russia, contexts in which shadow banking supply by non-financial enterprises tends to crowd out real investment. A higher real investment ratio ($\tilde{I}_r$) acts as a pull factor, drawing firms away from regulatory arbitrage and encouraging the reallocation of capital toward productive use. This substitution effect lies at the heart of Hypothesis H2b. Conversely, a higher financial investment ratio ($\tilde{I}_f$) reflects stronger firm-level financialisation and lower opportunity costs of engaging in shadow banking. Financial investment thus acts as a push factor into shadow banking, a mechanism formalised in Hypothesis H2a. While investment ratios may themselves be influenced by financing conditions, we address potential endogeneity concerns in the empirical analysis. Taken together, $\tilde{I}_r$ and $\tilde{I}_f$ represent distinct supply-side channels through which firms engage in shadow banking.

In terms of measurement, financial investment is defined as the ratio of financial assets to total assets, following Demir (2009), Xu and Xuan (2021), and Zhang and Zheng (2024). Financial assets include items such as trading financial assets, derivative financial assets, net loans and advances, assets available for sale, held-to-maturity investments, and investment properties. Real investment is measured as expenditures on fixed assets, intangible assets, and related capital formation relative to total assets, following Tang and Zhang (2019). These continuous measures capture firms' allocation between financial and productive activities. Financial and real investment orientation are inherently industry-specific: the level of financial or real investment that characterises an oriented firm differs across sectors. To account for this, we construct the industry-level asset ratio as the industry-year average of the respective investment ratio and use it as a benchmark against which each firm's own investment ratio is compared. The resulting indicators remain fully firm-specific and capture whether a firm is more or less financially (or real) oriented relative to its industry peers in a given year, thereby controlling for sector-wide differences in financial structure and investment intensity. Based on this benchmark, we transform the continuous measures into binary indicators. This transformation reduces sensitivity to outliers and measurement error in firm-level financial data and allows us to capture differences in firms' orientation rather than marginal variation in investment shares.

3.1.4. Other control variables

Shadow banking is subject to the investment and financing behavior of enterprises (Allen et al., 2019; Han and Li, 2020; Jiang et al., 2023). We therefore include a number of variables in our model that control for observable firm characteristics, firms' financial investment behaviour and environment. Considered control variables include firm size ($Size$), firm age (Age), leverage ratio (Lev), return on assets (ROA), revenue growth rate ($Growth$), and Tobin Q ($TobinQ$). We further control for institutional investor ownership (IO), the ownership ratio of the largest shareholder ($Top1$), and the degree of corporate financing constraints (FC) which could potentially influence managerial decisions regarding capital allocation.

The definitions and measurement of variables employed in this study are show in Table 1. Table 2 presents summary statistics of our main variables. Shadow banking ($\ln I_f^{SB}$) has a mean of 18.371 and a relatively large standard deviation of 2.415. There is therefore a large

difference in shadow banking activities between firms, that the range of the data further emphasises: The minimum and maximum values os $\ln I_f^{SB}$ are 12.405 and 23.448, respectively, reflecting a large spectrum of financialisation for considered firms. These summary statistics reveal considerable heterogeneity in shadow banking between firms, which is unsurprising, considering China's corporate landscape. We consider this in heterogeneity analyses below.

Bank competition (HHI) is on average 0.309 with a standard deviation of 0.073. The minimum and maximum values are 0.167 and 0.596, respectively. This implies a moderately concentrated level of bank concentration along cities considered, which is subject to relatively narrow dispersion. The range is again quite large, however, with the maximum being particularly high. This supports our approach to consider city-level indices as a benchmark as well as province-level ones for robustness.

Table 1
Variable definition.

Item	Symbol	Measurement	Data Source
Shadow Banking	I_f^{SB}	Sum of entrusted loans, entrusted financial management, private lending, bank financial management, securities financial management, trust products, and structured deposits.	CSMAR financial statements ("other current assets", "other receivables"), CSMAR external investment database
Shadow Banking	I_f^{EL}	Sum of entrusted loans.	CSMAR financial statements
Bank Competition	HHI	Index. See formula (1).	CBIRC bank branch license records (following Degryse and Ongena, 2007; Du and Cheng, 2022)
Financial Investment Ratio	$\tilde{I}_f$	(Trading financial assets + derivative financial assets + net loans + advances issued + net available for sale financial assets + net held-to-maturity investments + net investment real estate + debt investments + other debt investments + long-term equity investments + net long-term equity investments + other equity instruments investments) / total assets.	CSMAR
Real Investment Ratio	$\tilde{I}_r$	(Total cash flow paid for the purchase + construction of fixed assets + intangible assets + other long-term assets + net amount of construction in progress) / total assets.	CSMAR
Firm Size	$Size$	Logarithm of total assets.	CSMAR
Firm Age	Age	Logarithm of enterprise age.	CSMAR
Leverage Ratio	Lev	Total liabilities / total assets.	CSMAR
Return on Assets	ROA	Net profit / total assets.	CSMAR
Revenue Growth Rate	$Growth$	Annual growth rate of total sales.	CSMAR
Tobin Q	$TobinQ$	Sum of stock market value and total debt / total assets.	CSMAR, Wind
Institutional Investor Ownership	IO	Shares held by institutional investors / total capital.	CSMAR
Ownership Ratio of the Largest Shareholder	$Top1$	Shares held by the largest shareholder / total capital.	CSMAR
Corporate Financing Constraints	FC	$FC = 0.737 \times Size + 0.043 \times Size^2 - 0.040 \times Age$.	Constructed variable (based on CSMAR inputs)

Table 2
Descriptive statistics of variables.

Variable	Observations	Mean	SD	Min	Median	Max	Skewness
$\ln I_f^{SB}$	24,659	18.371	2.415	12.405	18.383	23.448	-0.122
HHI	24,659	0.309	0.073	0.167	0.309	0.596	0.613
$Size$	24,659	22.031	1.313	19.200	21.877	26.174	0.640
Age	24,659	2.788	0.366	1.609	2.833	3.466	-0.820
Lev	24,659	0.468	0.219	0.066	0.459	1.357	0.879
ROA	24,659	0.041	0.066	-0.288	0.041	0.211	-1.648
$Growth$	24,659	0.164	0.409	-0.555	0.102	2.710	3.281
$TobinQ$	24,659	2.004	1.356	0.865	1.574	9.022	2.791
IO	24,659	43.796	25.931	0.295	45.335	94.793	-0.047
$Top1$	24,659	34.028	15.447	1.000	32.080	74.220	0.356
FC	24,659	-3.785	0.259	-4.383	-3.786	-3.079	0.005

Notes: This table shows the summary statistics for the variables employed in our baseline regression. The sample consists of firms listed on the Shanghai and Shenzhen stock exchanges from 2011 to 2020.

3.2. Heterogeneity

An important part of our paper is to consider different dimensions of heterogeneity that might have an impact on results. These include individual firm-level characteristics, as well as industry specific effects that play a role in firms' capital allocation decisions and thus affect the channels we seek to describe. A key part in our paper is the distinction between SOEs and private firms as SOEs might face weaker borrowing constraints. These may again be industry-specific as Figure 1 below illustrates, where some sectors are almost completely dominated by SOEs whilst other are almost fully privately organised. In the figure, we plot SOE Shares in the total of firms (red) to give a sense of the relative importance of different industries in our data, where manufacturing industries represent the majority of firms, whilst health and education are negligible.

The industry-level heterogeneity is important for firms' investment behaviour as Figure 2 below illustrates: Particularly real investment ratios show large variation across industries—between above 20% for mining and below 5% in construction and wholesale. In addition to cross-industry variation, there is substantial within-industry variation in several industries. This highlights the importance of firm-level heterogeneity. We address both by including industry as well as firm-level fixed effects in our model and by investigating different sources of heterogeneity in an extensive heterogeneity analysis.

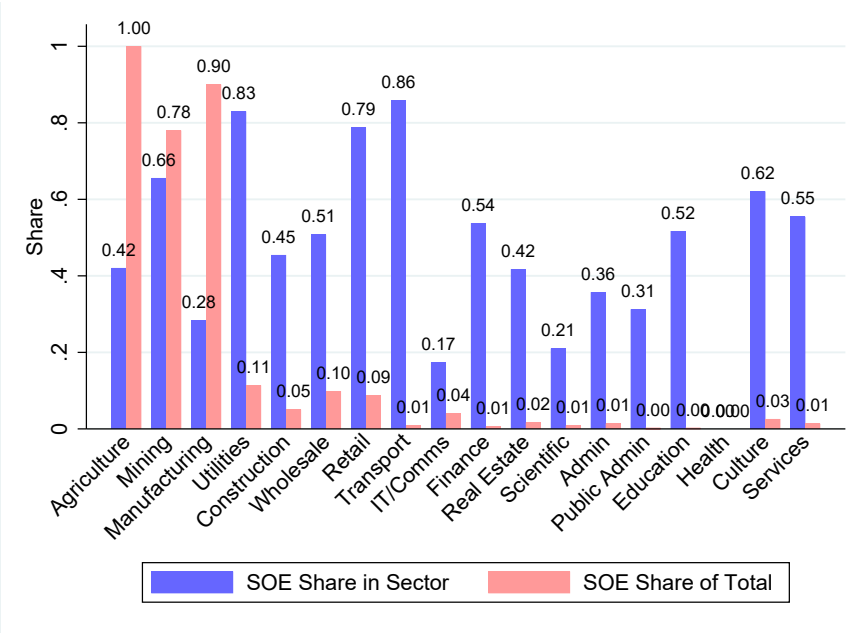


Fig 1. Share of state-owned enterprises (SOE) by aggregate industry sector, measured as the proportion of SOEs in total firm counts within each sector, alongside each sector's share in the overall sample.

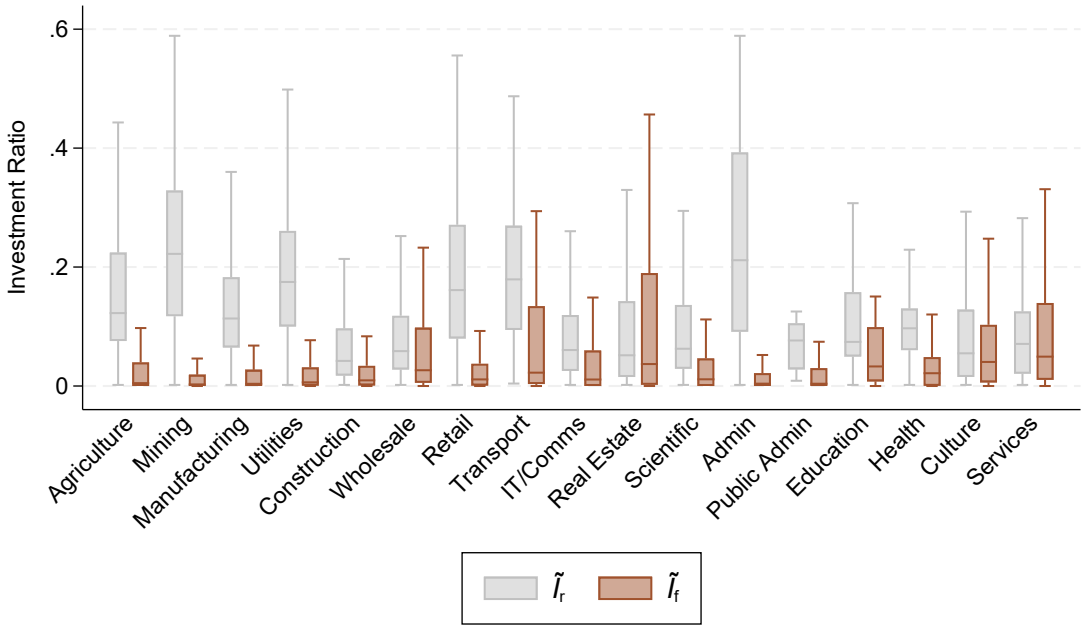


Fig 2. Distribution of firm-level investment ratios—both financial and real—within and across aggregate industry sectors. Each observation corresponds to a firm-level ratio; the figure illustrates the distribution across firms within each sector. See Section 3.1.3 for more details on measurement.

A further source of heterogeneity in our setting is geographical variation in firms' shadow banking activity. It is important as differences in local banking structures are closely linked to the intensity of bank competition and thereby firms' incentives to engage in shadow banking. Figure 3 illustrates the regional distribution of shadow banking activity across provinces. Shadow banking activity varies substantially across provinces, reflecting differences in local financial environments and banking structures. We account for this variation in our empirical design through city-level measures of bank competition and fixed effects.

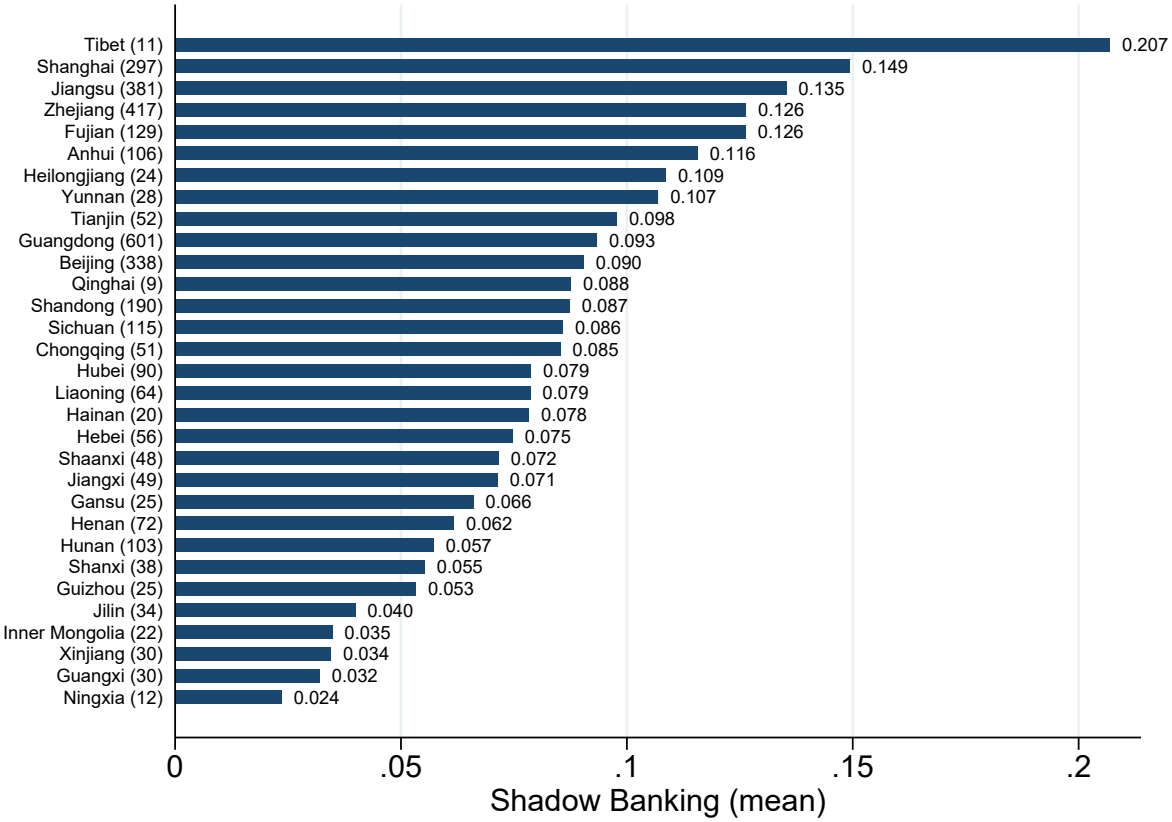


Fig 3. Shadow banking by province in 2020. Values represent the mean of firm-level shadow banking activity aggregated at the provincial level. See Section 3.1.1 for more details on measurement.

4. Empirical design and results

4.1. Impact of bank competition on shadow banking

To investigate the impact of bank competition on the shadow banking activities of non-financial firms, we estimate the following baseline specification:

$$\begin{aligned} \ln l^{SB}_{f_{i,c,j,t}} = \beta_0 + \beta_1 HHI_{c,t} + \beta_2 Z_{i,t} + \beta_3 (HHI_{c,t} \times Z_{i,t}) \\ + \boldsymbol{\delta}' \mathbf{X}_{i,t} + \gamma_t + \mu_{i,j} + \theta_i + \epsilon_{i,c,j,t}, \end{aligned} \tag{2}$$

where, $\ln l^{SB}_{f_{i,c,j,t}}$, is the logarithm of shadow banking for firm i in city c, industry j, and year t. $HHI_{c,t}$ is the Herfindahl-Hirschman Index measuring city-level bank concentration. $Z_{i,t}$ is a placeholder for either financial investment intensity ($Z = \tilde{I}_{f_{i,t}}$) or real investment intensity ($Z = \tilde{I}_{r_{i,t}}$),[9] depending on the specification. We estimate two versions of equation (2) separately to assess whether the effect of bank competition on shadow banking is moderated differently by firms' financial versus real investment behaviour. The interaction term captures how the effect of bank competition varies with firms' investment orientation. $\mathbf{X}_{i,t}$ is a vector of control variables. γ_t, $\mu_{i,j}$, and θ_i capture year, industry, and firm fixed effects (FE), respectively.

We first consider a specification that excludes moderating effects to examine hypothesis H1. Columns (1) and (2) of Table 3 show that bank competition enters positively but is statistically insignificant, albeit marginally so. Adding further control variables only slightly changes the effect size. A one-standard-deviation increase in bank concentration (0.073) is associated with an increase in shadow banking of approximately 4.2%. This suggests that an increase in bank concentration (i.e., a decrease in competition) alone does not significantly raise shadow banking activity. These findings caution against interpreting the effect of bank concentration on shadow banking in isolation and instead point to the role of moderating mechanisms—such as firms' investment orientation—that we examine in subsequent sections. This is consistent with evidence that the effects of financial conditions depend on firm characteristics and institutional context (Petersen and Rajan, 1995; Beck et al., 2006).

4.1.1. Corporate Real and Financial Activity

Our main objective is to test how firms' financial orientation $\tilde{I}_f$ and real investment intensity $\tilde{I}_r$ influence their shadow banking activity, and how these internal capital structures moderate the effect of bank concentration. Following the logic outlined in Section 2.2, we expect financialised firms to exhibit dampened responsiveness to competition (a weaker push effect), while real-intensive firms should respond more strongly as constraints ease (a stronger pull effect). To capture these heterogeneous responses, we estimate interaction terms between bank concentration and both investment measures in equation (2).

We primarily focus on the coefficient β_3, which captures how the marginal effect of bank concentration varies with firms' investment orientation. When $Z = \tilde{I}_{f_{i,t}}$, a positive β_3 indicates that the effect of bank concentration on shadow banking becomes less negative as financialisation increases—consistent with the dampening of competition's disciplining effect for highly financialised firms, in line with Hypothesis H2a. When $Z = \tilde{I}_{r_{i,t}}$, a similarly positive coefficient does not contradict Hypothesis H2b: depending on the baseline effect, it can reflect a stronger marginal response among real-intensive firms. As discussed earlier, this interpretation hinges on the level effect of bank concentration (β_1), which determines whether

[9]The financial investment ratio is measured by the proportion of financial assets to total assets and real investment is measured by the proportion of physical assets to total assets. We transform both measures into a binary indicator that equals one when an industry's respective asset ratio surpass the annual sample mean, and zero otherwise.

a given moderation amplifies or attenuates the total effect. The interaction coefficient β_3 thus needs to be evaluated jointly with β_1, as the same sign on β_3 may reflect different underlying mechanisms depending on the direction and strength of the baseline relationship.

Columns (3) and (4) of Table 3 report the results after introducing the enterprise financialization ratio, with relevant variables centered. The coefficients of the interaction term, $HHI \times \tilde{I}_f$, are significantly positive, indicating that the marginal effect of bank competition becomes weaker for firms with higher financial investment—consistent with H2a. The negative and statistically significant coefficient on $\tilde{I}_f$ suggests that more financialised firms are less likely to expand shadow banking activity. In economic terms, this moderating effect is substantial: using the specification in Column (4), a one-standard-deviation increase in bank concentration implies an increase in shadow banking of about 3.2% for firms with low financial orientation, compared to approximately 78.1% for highly financialised firms. Note that the magnitude of this effect reflects how the effect of a one-standard-deviation increase in bank concentration differs for firms with relatively higher financial orientation compared to their industry benchmark, and not marginal changes in investment orientation. This notwithstanding, it reveals a important role of investment orientation in moderating the effect of bank concentration on shadow banking. This supports our theoretical argument that these firms operate further along the concave financial return function, where marginal returns to additional financial allocation are diminished. As a result, they exhibit a weaker inclination to intensify shadow banking, consistent with the dampened push effect described in Hypothesis H2a. Overall, these results suggest that while bank competition tends to reduce enterprise shadow banking, its effect is moderated by firms' capital structure: the disciplining impact is weaker for more financialised firms, consistent with diminishing marginal returns and a dampened push channel.

Results for real instead of financial investment ratios are shown in Columns (5) and (6) of Table 3. The interaction term $HHI \times \tilde{I}_r$ is again positive and statistically significant. Given that the estimated coefficient on HHI is negative, this implies that the marginal effect of bank concentration on shadow banking becomes more negative—that is, more disciplining—as real investment intensity increases. Specifically, a one-standard-deviation increase in bank concentration is associated with a decrease in shadow banking of about 2.5% for firms with low real investment, but an increase of approximately 36.7% for firms with high real investment. This supports Hypothesis H2b: capital-intensive firms respond more strongly to changes in banking market structure, reallocating away from shadow banking when competition intensifies. Notably, the coefficient on $\tilde{I}_r$ is negative, but its magnitude is considerably smaller than the coefficient on $\tilde{I}_f$ in the previous specification. Since both variables are measured on the same scale, this suggests that baseline differences in real orientation are less predictive of shadow banking than baseline financial orientation—consistent with the idea that real investment acts more as a moderator (pull factor) than as a direct driver. These results confirm the role of real investment as a reinforcing factor that strengthens the disciplining effect of bank competition on shadow banking engagement.

Table 3
Baseline results.

	(1) $\ln I_f^{SB}$	(2) $\ln I_f^{SB}$	(3) $\ln I_f^{SB}$	(4) $\ln I_f^{SB}$	(5) $\ln I_f^{SB}$	(6) $\ln I_f^{SB}$
HHI	0.808	0.715	0.707	0.599	0.026	-0.268
	(0.626)	(0.577)	(0.632)	(0.582)	(0.737)	(0.679)
$\tilde{I}_f$			-1.512	-1.603		
			(1.469)	(1.412)		
$\tilde{I}_r$					-1.024	-1.557**
					(0.718)	(0.659)
$HHI \times \tilde{I}_f$			6.889	8.001*		
			(4.778)	(4.526)		
$HHI \times \tilde{I}_r$					4.517**	5.715***
					(2.149)	(1.968)
Controls	NO	YES	NO	YES	NO	YES
Firm FE	YES	YES	YES	YES	YES	YES
Industry-Year FE	YES	YES	YES	YES	YES	YES
Observations	24138	24138	24138	24138	24138	24138
Adjusted R^2	0.613	0.639	0.614	0.639	0.614	0.639

Notes: The dependent variable is corporate shadow banking activity, measured as $\ln I_f^{SB}$. HHI denotes the city-level Herfindahl—Hirschman Index of bank concentration. Columns (1)–(2) report specifications without interaction terms. Columns (3)–(4) include interaction terms between HHI and financial investment intensity ($\tilde{I}_f$), while Columns (5)–(6) include interaction terms between HHI and real investment intensity ($\tilde{I}_r$). Control variables comprise $Size$, Age, Lev, ROA, $Growth$, $TobinQ$, IO, $Top1$, and FC. Control variables are included consistently across specifications with controls, but their coefficients are not reported to preserve readability. All regressions include firm and industry-by-year fixed effects. Standard errors clustered at the firm level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels. Detailed variable definitions and data sources are provided in Table 1.

4.2. Robustness checks

To assess the reliability of our main findings, we conduct several robustness exercises that address measurement concerns, omitted variable bias, and endogeneity. First, we verify that our results are not driven by the specific construction of the bank competition and shadow banking variables. While our baseline specification uses the Herfindahl-Hirschman Index at the city level (HHI) as the core explanatory variable, we additionally employ alternative measures of bank competition, the provincial-level HHI index (HHI^p). As shown in Table B.1 of Appendix B, the results remain consistent in direction across all specifications, supporting the robustness of our conclusions with respect to how bank competition affects shadow banking behavior. We also verify the robustness of our outcome variable by replacing the logarithm of the shadow banking ratio with the logarithm of entrusted loans from non-financial firms ($\ln I_f^{EL}$), a narrower and arguably more conservative measure of shadow banking supply.

We approach potential endogeneity concerns using a combination of complementary empirical strategies in line with common practice in the literature (Petersen, 2009). In the

absence of a natural or quasi-experimental setting, this involves assessing the robustness of the results across specifications that address different sources of bias. Specifically, we employ instrumental variable estimation to isolate plausibly exogenous variation in bank competition, dynamic panel methods to account for persistence and potential feedback effects, and alternative specifications to test sensitivity to measurement and specification choices. The consistency of results across these approaches indicates that the findings are not driven by a particular specification or source of variation. While these approaches mitigate key concerns, the results should be interpreted as evidence on robust conditional relationships rather than definitive causal effects.

To address endogeneity arising from potential reverse causality we employ a two-stage instrumental variable (IV) approach using lagged values of bank concentration and the corresponding lagged interaction terms as instruments. Specifically, lagged bank concentration is used as an instrument for contemporaneous HHI, while the interaction specifications instrument $HHI \times \tilde{I}_f$ and $HHI \times \tilde{I}_r$ using the corresponding lagged interaction terms. This approach exploits the persistence in local banking market structure and firms' investment orientation while relying on predetermined variation that is plausibly exogenous to contemporaneous shadow banking activity. The first-stage results, reported in Table 4, show that the instruments are strongly correlated with the endogenous regressors, with Kleibergen-Paap F-statistics well above conventional thresholds. The second-stage regressions continue to support our baseline results, indicating that the observed relationship between bank competition and shadow banking activity is unlikely to be driven by simultaneity bias. In particular, the interaction effects remain consistent with the theoretical push- and pull-channel mechanisms developed in Section 2.2.

Table 4

IV estimates: Instrumenting HHI and its interactions.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Instrumental variable	iv_HHI		iv_HHI	$iv_HHI \times \tilde{I}_f$		iv_HHI	$iv_HHI \times \tilde{I}_r$	
Regression stage (Dept. variable)	First (HHI)	2SLS ($\ln I_f^{SB}$)	First (HHI)	First ($HHI \times \tilde{I}_f$)	2SLS ($\ln I_f^{SB}$)	First (HHI)	First ($HHI \times \tilde{I}_r$)	2SLS ($\ln I_f^{SB}$)
iv_HHI	0.494***		0.498***	-0.007***		0.478***	-0.061***	
	(0.025)		(0.025)	(0.001)		(0.026)	(0.004)	
HHI		0.141			-0.033			-0.710
		(1.300)			(1.302)			(1.389)
$(iv_HHI) \times \tilde{I}_f$			-0.193**	0.861***				
			(0.095)	(0.032)				
$\tilde{I}_f$			0.058**	0.036***	-2.156			
			(0.028)	(0.009)	(1.663)			
$HHI \times \tilde{I}_f$					8.397			
					(5.396)			
$(iv_HHI) \times \tilde{I}_r$						0.099**	0.890***	
						(0.042)	(0.017)	
$\tilde{I}_r$						-0.030**	0.026***	-1.632**
						(0.012)	(0.005)	(0.812)
$HHI \times \tilde{I}_r$								5.085**
								(2.507)
Constant	0.152***		0.150***	0.003**		0.156***	0.016***	
	(0.020)		(0.020)	(0.001)		(0.020)	(0.003)	
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Industry-Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Observations	20541	20541	20541	20541	20541	20541	20541	20541
Adjusted R^2	0.959	0.031	0.959	0.997	0.032	0.959	0.993	0.031
F-statistics (P-value)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Kleibergen-Paap rk LM (P-value)		0.000			0.000			0.000
Kleibergen-Paap Wald rk F		405.46			204.03			200.11

Notes: This table reports the results of the two-stage instrumental variable regression. The instrumental variables are listed in the second row: iv_HHI for Column (1), (3) and (6), $iv_HHI \times \tilde{I}_f$ for Columns (4)–(5), and $iv_HHI \times \tilde{I}_r$ for Columns (7)–(8). Columns (1), (3), (4), (6), and (7) present the first-stage regression results, where the instrumental variables are regressed on the corresponding endogenous variables (HHI, $HHI \times \tilde{I}_f$, or $HHI \times \tilde{I}_r$). Columns (2), (5), and (8) present the second-stage results on the dependent variable ($\ln I_f^{SB}$). Kleibergen-Paap F statistics are all significant at the 10% level. Control variables comprise $Size$, Age, Lev, ROA, $Growth$, $TobinQ$, IO, $Top1$, and FC. Control variables are included consistently across specifications, but their coefficients are not reported to preserve readability. All regressions include firm and industry-by-year fixed effects. Standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1% levels, respectively. Detailed definitions of variables and their data sources can be found in Table 1.

Table 5
IV estimates: Instrumenting $\tilde{I}_f$, $\tilde{I}_r$, and HHI interactions.

	(1)	(2)	(3)	(4)	(5)	(6)
Instrumental variable	$iv_\tilde{I}_f$	$HHI \times iv_\tilde{I}_f$		$iv_\tilde{I}_r$	$HHI \times iv_\tilde{I}_r$	
Regression stage (Dept. variable)	First ($\tilde{I}_f$)	First ($HHI \times \tilde{I}_f$)	2SLS ($\ln I_f^{SB}$)	First ($\tilde{I}_r$)	First ($HHI \times \tilde{I}_r$)	2SLS ($\ln I_f^{SB}$)
$iv_\tilde{I}_f$	0.005***	-0.001*				
	(0.002)	(0.001)				
$HHI \times iv_\tilde{I}_f$	0.007	0.011***				
	(0.005)	(0.002)				
HHI	-0.104	-0.167***	0.412	-0.468*	-0.304***	-5.085**
	(0.100)	(0.034)	(0.898)	(0.255)	(0.088)	(2.237)
$\tilde{I}_f$			-6.104			
			(5.812)			
$HHI \times \tilde{I}_f$			24.809			
			(17.710)			
$iv_\tilde{I}_r$				0.005	-0.004***	
				(0.004)	(0.001)	
$HHI \times iv_\tilde{I}_r$				0.027*	0.026***	
				(0.014)	(0.005)	
$\tilde{I}_r$						-14.007***
						(3.971)
$HHI \times \tilde{I}_r$						35.568***
						(11.979)
Constant	0.104	0.077***		0.356***	0.163***	
	(0.081)	(0.026)		(0.122)	(0.038)	
Controls	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Industry-Year FE	YES	YES	YES	YES	YES	YES
Observations	20541	20541	20541	20541	20541	20541
Adjusted R^2	0.589	0.596	0.029	0.596	0.603	-0.008
F-statistics (P-value)	0.000		0.000	0.000	0.000	0.000
Kleibergen-Paap rk LM (P-value)			0.000			0.000
Kleibergen-Paap Wald rk F			104.44			78.86

Notes: This table reports the results of the two-stage instrumental variable regression. The instrumental variables are listed in the second row: $iv_\tilde{I}_f$ for Column (1), $HHI \times iv_\tilde{I}_f$ for Columns (2) and (3), $iv_\tilde{I}_r$ for Column (4), and $HHI \times iv_\tilde{I}_r$ for columns (5) and (6). Columns (1), (2), (4) and (5) present the first-stage regression results, where the instrumental variables are regressed on the corresponding endogenous variables (HHI, $HHI \times \tilde{I}_f$, or $HHI \times \tilde{I}_r$). Columns (3) and (6) present the second-stage results on the dependent variable ($\ln I_f^{SB}$). Kleibergen-Paap F statistics are all significant at the 10% level. Control variables comprise $Size$, Age, Lev, ROA, $Growth$, $TobinQ$, IO, $Top1$, and FC. Control variables are included consistently across specifications, but their coefficients are not reported to preserve readability. All regressions include firm and industry-by-year fixed effects. Standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1% levels, respectively. Detailed definitions of variables and their data sources can be found in Table 1.

Table 5 gives estimates obtained from using instruments for $\tilde{I}_f$ and $\tilde{I}_r$ and their interaction terms respectively. As shown in Columns (3) and (6) of Table 5, both instruments pass the weak instrument test, and the resulting coefficient estimates remain in line with those of the baseline regressions, reinforcing our interpretation of $\tilde{I}_f$ and $\tilde{I}_r$ as supply-side channels of shadow banking. To explore dynamic interactions among our key variables, we further estimate a panel vector autoregression (PVAR) model. This approach allows us to treat all variables as endogenous and capture their lagged relationships. The impulse response functions, presented in Figures B.1 and B.2 of Appendix B, reveal that a positive shock

to HHI—indicating a decline in bank competition—leads to a significant and persistent increase in shadow banking. Conversely, the response of HHI to shadow banking activity is comparatively modest, consistent with our main causal direction. Additionally, shocks to $\tilde{I}_f$ increase shadow banking activity, while shocks to $\tilde{I}_r$ have a negative effect, albeit with wider confidence intervals. These dynamic results corroborate the baseline regression analysis and support our interpretation of financial and real investment as push and pull factors in firms' shadow banking supply decisions.

Taken together, these robustness exercises strengthen the credibility of our empirical findings and demonstrate that the observed relationships are not artifacts of particular measurement choices or estimation strategies.

5. Heterogeneity analysis

Building on the baseline regressions and interaction effects presented in Section 4, we now turn to firm- and environment-specific heterogeneity that may shape how bank competition influences shadow banking. Given the institutional features discussed in Section 2—such as credit access asymmetries, governance quality, and regulatory arbitrage—we examine whether the disciplining effect of bank competition is conditional on variation in external oversight, ownership structure, and corporate governance. To assess whether apparent subgroup differences reflect statistically meaningful heterogeneity, we complement the subgroup regressions with bootstrap coefficient-difference tests for the interaction terms.

5.1. Regulatory environment and oversight

Regulatory arbitrage enables non-financial enterprises to circumvent formal banking regulations by engaging in credit intermediation through unregulated channels such as shadow banking. In the Chinese context, entrusted loans serve as a prominent example: non-financial firms extend credit to one another, effectively substituting for banks while avoiding the capital adequacy, disclosure, and risk management requirements imposed on regulated financial institutions. This circumvention erodes market transparency and exacerbates information asymmetries among banks, enterprises, and regulators, as highlighted by Allen et al. (2019). The lack of supervision over these informal credit flows can elevate financial risk and systemic fragility, particularly when shadow banking operates outside prudential oversight (Ryan et al., 2014). Although enhancing external regulatory standards may help mitigate information asymmetries and reduce financial risk, such measures must directly address the loopholes that enable arbitrage. Otherwise, tighter regulation of the formal banking sector may unintentionally accelerate the growth of shadow banking. Recent evidence supports this dynamic: stricter regulation has been shown to curb the financialization of non-financial firms and limit their engagement in shadow banking (Yang et al., 2019; Zhang et al., 2023). Therefore, it is reasonable to expect that under different external regulatory environments, the relationship between bank competition and shadow banking activity among non-financial enterprises is heterogeneous and shaped by the extent to which arbitrage opportunities are available.

To examine how external regulatory conditions affect the relationship between bank competition and shadow banking, we follow Liu (2011) and construct a variable for external regulatory attention, Analyst, based on analyst coverage. Firms with above-industry-average analyst coverage in a given year are classified as having high regulatory attention ($Analyst_1$), while those below the average are classified as having low attention ($Analyst_0$).[10] The heterogeneity regression results, presented in Table 6, show that the coefficient is statistically significant only for the high-attention group, suggesting that stronger external oversight may reinforce the disciplining effect of bank competition on shadow banking. The interaction effects are positive and statistically significant in both analyst-coverage subsamples, with larger coefficients in the high-coverage group. Bootstrap coefficient-difference tests indicate significant heterogeneity across analyst-coverage groups only in the real-investment channel. The corresponding p-values are 0.109 for $HHI \times \tilde{I}_f$ and 0.126 for $HHI \times \tilde{I}_r$.

5.2. Firm ownership

China's shadow banking landscape is shaped by institutional asymmetries between state-owned enterprises (SOEs) and private firms. SOEs, benefiting from preferential access to formal credit and implicit government support, often act as key suppliers of shadow banking credit, profiting from interest rate differentials between formal and informal channels. Intensified bank competition disrupts this dynamic by expanding credit access to private enterprises and eroding SOEs' arbitrage opportunities, thus curbing their incentive and capacity to supply shadow credit.[11]. However, SOEs tend to be more financialised than private firms, meaning they operate further along the concave segment of the financial return function, where marginal returns diminish. As a result, the effect of competition on reallocating capital away from shadow banking is weaker for financialised SOEs, consistent with a dampened push-channel response. At the same time, SOEs are more responsive to regulatory priorities due to policy alignment pressures and operate under softer budget constraints (Brandt and Li, 2003; Allen et al., 2005; Jiang et al., 2020). As competition grows, they are more likely to reallocate resources toward the real economy in line with regulatory aims. In contrast, private firms—less tied to state policy and more reliant on informal finance—respond to competition primarily through improved credit access, facing weaker institutional incentives to reduce shadow banking engagement (Ge et al., 2012; Malmendier and Tate, 2015).

To assess how ownership structure shapes the relationship between bank competition and shadow banking, we divide firms into state-owned (SOE_1) and private (SOE_0) groups and estimate interaction terms for both financial investment ($\tilde{I}_f$) and real investment intensity ($\tilde{I}_r$). The interaction terms are again specified using centered variables with full interaction notation, meaning the reported coefficients capture marginal effects at the mean of the interacting variables rather than simple product terms. This approach helps mitigate multicollinearity by partialling out the portion of the variation in the interaction term

[10]Following Liu (2011), firms with above-average analyst coverage are considered to have high regulatory attention because analysts act as external monitors, enhancing transparency and discipline in ways comparable to formal oversight.

[11]For instance, SOEs have been documented to channel funds to private real estate developers via entrusted loans, a practice prevalent in China's shadow banking system (see Tian et al. (2024))

Table 6
Regulatory environment and oversight.

	(1)	(2)	(3)	(4)
	Financial investment		Real investment	
	$Analyst_0$	$Analyst_1$	$Analyst_0$	$Analyst_1$
HHI	1.178*	-1.129	0.516	-2.666**
	(0.713)	(1.156)	(0.823)	(1.322)
$\tilde{I}_f$	-2.240	-5.477		
	(1.589)	(3.556)		
$\tilde{I}_r$			-1.231	-2.780**
			(0.777)	(1.251)
$HHI \times \tilde{I}_f$	8.910*	20.278*		
	(5.132)	(10.992)		
$HHI \times \tilde{I}_r$			4.943**	9.641***
			(2.316)	(3.716)
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Industry-Year FE	YES	YES	YES	YES
Observations	15481	7830	15481	7830
Adjusted R^2	0.620	0.678	0.620	0.678

Notes: This table reports the results of heterogeneity analysis regarding enterprises' external oversight (*Analyst*). The findings indicate that the restraining effect of bank competition on corporate shadow banking activities is more pronounced when external oversight is weaker. Bootstrap coefficient-difference tests indicate statistically significant heterogeneity across analyst-coverage groups in the real-investment channel, but not in the financial-investment channel. Control variables comprise *Size*, *Age*, *Lev*, *ROA*, *Growth*, *TobinQ*, *IO*, *Top*1, and *FC*. Control variables are included consistently across specifications with controls, but their coefficients are not reported to preserve readability. All regressions include firm and industry-by-year fixed effects. Standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1% levels, respectively. Detailed definitions of variables and their data sources can be found in Table 1.

that arises purely from the correlation between the constituent variables, isolating the true moderating effect. If bank competition constrains SOEs' shadow banking activity more effectively, we expect a more positive interaction coefficient on $HHI \times \tilde{I}_r$ for SOEs, while we expect dampened (less positive) coefficient on $HHI \times \tilde{I}_f$, as due to the higher degree of financialisation implying that firms operate on the diminishing end of the financial return function (as discussed for baseline results).

Table 7 presents the resulting estimates. Columns (1) and (2) focus on the interaction between bank concentration and financial investment across ownership types. The interaction term $HHI \times \tilde{I}_f$ is positive in both ownership subsamples and statistically significant only among private firms. While this pattern suggests a somewhat stronger financial-channel response among private firms, coefficient-difference tests do not reject equality of the interaction coefficients across ownership groups ($p = 0.379$). We therefore interpret the difference as suggestive rather than statistically conclusive. For the real-investment channel, the interaction term $HHI \times \tilde{I}_r$ is economically larger among SOEs. However, coefficient-difference

Table 7
Firm ownership.

	(1)	(2)	(3)	(4)
	Financial investment		Real investment	
	SOE_0	SOE_1	SOE_0	SOE_1
HHI	0.552	0.156	-0.341	-1.353
	(0.705)	(1.019)	(0.800)	(1.223)
$\tilde{I}_f$	-3.125*	-1.634		
	(1.645)	(2.549)		
$\tilde{I}_r$			-1.417*	-3.238***
			(0.795)	(1.148)
$HHI \times \tilde{I}_f$	11.571**	9.104		
	(5.208)	(8.663)		
$HHI \times \tilde{I}_r$			6.052**	9.418***
			(2.392)	(3.423)
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Industry-Year FE	YES	YES	YES	YES
Observations	15561	8377	15561	8377
Adjusted R^2	0.618	0.698	0.618	0.698

Notes: This table reports the results of heterogeneity analysis regarding enterprises' property rights nature (SOE). The findings indicate that the restraining effect of bank competition on corporate shadow banking activities is more pronounced for non-state-owned enterprises. Coefficient-difference tests show that the financial-channel interaction does not differ significantly across ownership groups, whereas the real-investment interaction is economically stronger among SOEs, although the difference is only marginally insignificant statistically. Control variables comprise $Size$, Age, Lev, ROA, $Growth$, $TobinQ$, IO, $Top1$, and FC. Control variables are included consistently across specifications with controls, but their coefficients are not reported to preserve readability. All regressions include firm and industry-by-year fixed effects. Standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1% levels, respectively. Detailed definitions of variables and their data sources can be found in Table 1.

tests do not indicate statistically significant differences across ownership groups ($p = 0.197$), although the result is only marginally insignificant. We therefore interpret this pattern as economically suggestive rather than statistically conclusive. These results underscore the importance of internal capital structure in moderating competitive effects, particularly among state-owned firms.

5.3. Firm governance

CEO duality—when one individual serves as both CEO and board chair—is widely regarded as a corporate governance weakness. This concentration of power reduces board oversight and can intensify agency problems, increasing the likelihood of decisions misaligned with shareholder interests — governance problems that may also shape how firms respond to external pressures such as increased bank competition (Jensen and Meckling, 1976; Fama and Jensen, 1983). In firms with CEO duality, executives operate with fewer internal constraints, granting them greater discretion over strategic financial choices, includ-

ing engagement in shadow banking activities. Competitive banking markets can introduce external monitoring pressure that partially compensates for weak internal governance (Boot and Thakor, 2000). By imposing stricter lending standards and enhancing scrutiny of financial decisions, bank competition may discourage excessive reliance on shadow banking. Moreover, greater credit availability under competition may offer more attractive alternatives for real-sector investment, further reducing incentives to channel funds into shadow banking. We therefore expect that the disciplining effect of bank competition on shadow banking is stronger in firms with CEO duality, where external market forces substitute for limited internal oversight.

To examine whether the impact of bank competition on shadow banking is conditioned by internal governance, we split the sample by CEO duality ($Dual$), where $Dual = 1$ indicates the same individual holds both CEO and chair positions, a common proxy for weaker governance. Table 8 reports the results. The subgroup coefficients are somewhat larger in the CEO-duality subsample, particularly for the financial channel. However, coefficient-difference tests indicate that the interaction coefficients do not differ significantly across governance groups in either channel. The corresponding p-values are 0.332 for $HHI \times \tilde{I}_f$ and 0.345 for $HHI \times \tilde{I}_r$. The results therefore suggest only limited evidence that internal governance conditions systematically modify the moderating role of bank competition.

Table 8
Firm governance.

	(1)	(2)	(3)	(4)
	Financial investment		Real investment	
	$Dual_0$	$Dual_1$	$Dual_0$	$Dual_1$
HHI	1.456	0.056	0.101	-0.865
	(1.429)	(0.660)	(1.520)	(0.780)
$\tilde{I}_f$	-1.582	-2.154		
	(2.961)	(1.525)		
$\tilde{I}_r$			-1.590	-2.036***
			(1.339)	(0.761)
$HHI \times \tilde{I}_f$	6.020	9.424*		
	(9.470)	(4.839)		
$HHI \times \tilde{I}_r$			7.865*	6.369***
			(4.042)	(2.249)
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Industry-Year FE	YES	YES	YES	YES
Observations	7049	16509	7049	16509
Adjusted R^2	0.673	0.646	0.674	0.646

Notes: This table reports the results of heterogeneity analysis regarding enterprises' internal governance ($Dual$). The findings indicate that the restraining effect of bank competition on corporate shadow banking activities is more pronounced when internal governance is weaker. Coefficient-difference tests indicate that the interaction terms do not differ significantly across CEO-duality subsamples. Control variables comprise $Size$, Age, Lev, ROA, $Growth$, $TobinQ$, IO, $Top1$, and FC. Control variables are included consistently across specifications with controls, but their coefficients are not reported to preserve readability. All regressions include firm and industry-by-year fixed effects. Standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1% levels, respectively. Detailed definitions of variables and their data sources can be found in Table 1.

6. Discussion

The empirical results provide consistent evidence that the effect of bank competition on firms' shadow banking activity is conditional rather than uniform. In the baseline specification, increased bank competition is associated with a reduction in firm-level shadow banking. This is consistent with the view that stronger competition compresses intermediation margins and relaxes financing constraints, thereby reducing the incentives for firms to channel funds into shadow banking activities (Jiang et al., 2023; Wang et al., 2023).

However, this average effect masks substantial heterogeneity. The interaction results show that firms with higher financial orientation respond differently to changes in bank competition. For these firms, the sensitivity of shadow banking activity to competition is significantly attenuated. This suggests that financially oriented firms operate further along the intermediation return function and are therefore less responsive to marginal changes in competition. In contrast, firms with lower financial exposure exhibit a stronger response, indicating that competition primarily affects those firms at the margin of entering or expanding shadow banking activities. These patterns suggest that the effect of bank competition

operates primarily through firm-level margins rather than uniformly across firms. In this sense, the mixed evidence in the literature can be understood as reflecting differences in the composition of firms and their exposure to financial activities (Petersen and Rajan, 1995; Beck et al., 2006). Our results help reconcile these findings by showing that competition influences not only access to finance but also firms' allocation of resources between productive investment and financial intermediation.

The heterogeneity analysis further underscores the importance of institutional and organisational features. Differences between SOEs and non-SOEs, as well as variation in governance and information environments, indicate that access to finance and monitoring conditions continue to shape firms' responses to bank competition. These patterns are consistent with evidence that China's financial system remains segmented, with preferential access to credit for certain firms and continued reliance on alternative financing channels (Allen et al., 2019; Brandt and Li, 2003).

Our results hence suggest that bank competition affects firms' behaviour through two interconnected channels: by altering borrowing conditions and by changing the relative profitability of financial intermediation. The interaction between these channels generates the heterogeneous responses observed in the data and aligns with the mechanism emphasised in our theoretical framework.

At the same time, several caveats are warranted. Although we employ multiple empirical strategies to address endogeneity—including instrumental variable estimation and dynamic panel approaches—causal interpretation ultimately relies on identifying assumptions that cannot be fully tested. Moreover, our empirical framework abstracts from broader general equilibrium effects that may arise through interactions between firms, banks, and policy. These considerations suggest that the results should be interpreted as evidence on firm-level behavioural responses rather than as a complete account of the aggregate effects of bank competition.

7. Summary and conclusions

We investigate the impact of bank competition on shadow banking activities of non-financial firms in China, focusing on the internal reallocation of capital between real and financial uses. Using a stylised theoretical framework, we demonstrate how increased bank competition disincentivises firms' shadow banking supply and incentivises firms' real activity by relaxing borrowing constraints particularly for private enterprises, thereby shifting investment away from financialised assets and toward productive capital. Our empirical analysis supports this mechanism. However, in our setup, where firm and industry heterogeneity is accounted for, the direct effect of bank competition becomes statistically insignificant. This underscores the importance of moderating channels through firms' capital structure and internal characteristics in shaping shadow banking behaviour. We further consider asymmetry in borrowing constraints between SOEs and private firms that leads to constraints being more binding for the latter. An extensive analysis of firm heterogeneity plays a critical role in our empirical evidence in this regard: SOEs, which face softer budget constraints, and firms with weaker governance structures (e.g., CEO duality) exhibit systematically different

responses. In addition, we find that external regulatory attention, proxied by analyst coverage, moderates the disciplining effect of bank competition—firms with less external oversight respond more strongly, meaning that in the absence of external monitoring, market forces such as intensified bank competition play a more pronounced role in shaping firms' financial behaviour. A range of robustness exercises affirms these patterns.

The findings highlight the allocative implications of bank competition in credit-constrained environments. Rather than viewing shadow banking solely as a response to external financing gaps, our results show how internal investment decisions, shaped by institutional frictions, contribute to financialisation. In this context, SOEs—due to their preferential credit access—play a dual role: they benefit from relaxed constraints and simultaneously divert resources into the shadow banking system, particularly through entrusted loans and related instruments. In contrast, private firms appear more disciplined in reallocating capital toward real investments when credit conditions improve.

These dynamics carry clear policy implications. While promoting bank competition can ease financing constraints and reduce distortions in capital allocation, its effectiveness is mediated by ownership structures. SOEs' continued role as major suppliers of shadow credit weakens the disciplinary impact of market forces and sustains inefficient financial activity. This raises broader questions about the design of China's financial system: if SOEs are effectively acting as extensions of the banking sector, particularly under policy mandates, formalising that role—through clear regulatory oversight or institutional restructuring—may enhance transparency and reduce systemic risk. At the same time, strengthening corporate governance and ensuring a level playing field across ownership types would improve capital allocation efficiency.

While our study provides new insights into the interaction between banking structure and firm-level financial behaviour, it does not account for every possible channel or period-specific shock. In particular, we limit our sample to 2011–2020 to avoid structural disruptions from the 2008-2010 stimulus and the COVID-19 crisis. Given our data, there is also limited consideration of regulatory reforms to the sector, that were implemented gradually between 2018 and 2020. Future research could explore how alternative financial reforms or shifting regulatory regimes influence these dynamics in the longer run. Nonetheless, our results point to the need for both financial and corporate reform to curb distortive financialisation and support sustainable economic growth.

Appendix A. Proofs of hypotheses in Section 2

Appendix A.1. Hypothesis H1

We prove Hypothesis H1 by analyzing how bank competition affects the optimal allocation of capital between real and financial uses in a representative firm.

The firm receives capital endowment R, which it allocates between real investment I_r and financial investment I_f, subject to the constraint $I_r + I_f = R$. It solves

$$\max_{I_r, I_f} \quad \phi(I_r) + \theta(I_f; \kappa) \quad \text{s.t.} \quad I_r \leq \lambda(\kappa),$$

where $\phi' > 0$, $\phi'' < 0$, $\theta' > 0$, $\theta'' < 0$, $\lambda'(\kappa) > 0$, and $\partial\theta'/\partial\kappa < 0$. Here κ denotes the degree of bank competition: higher κ implies more competitive markets.

We consider two cases. First, suppose the borrowing constraint is slack, so $I_r < \lambda(\kappa)$. Then the optimal allocation satisfies the first-order condition,

$$\phi'(I_r^*) = \theta'(R - I_r^*; \kappa).$$

Differentiating both sides with respect to κ yields

$$\phi''(I_r^*) \cdot \frac{dI_r^*}{d\kappa} = -\theta''(I_f^*) \cdot \left(-\frac{dI_r^*}{d\kappa}\right) + \frac{\partial\theta'}{\partial\kappa},$$

which simplifies to

$$\left[\phi''(I_r^*) + \theta''(I_f^*)\right] \cdot \frac{dI_r^*}{d\kappa} = \frac{\partial\theta'}{\partial\kappa}.$$

Since the second derivatives are negative and $\partial\theta'/\partial\kappa < 0$, the right-hand side is negative while the denominator is also negative. Therefore,

$$\frac{dI_r^*}{d\kappa} > 0, \quad \frac{dI_f^*}{d\kappa} = -\frac{dI_r^*}{d\kappa} < 0.$$

That is, greater bank competition increases real investment and reduces financial investment. Second, suppose the borrowing constraint binds, so $I_r^* = \lambda(\kappa)$ and $I_f^* = R - \lambda(\kappa)$. Then

$$\frac{dI_r^*}{d\kappa} = \lambda'(\kappa) > 0, \quad \frac{dI_f^*}{d\kappa} = -\lambda'(\kappa) < 0.$$

Again, competition reallocates capital from financial to real uses.

In both cases, an increase in bank competition ($\kappa \uparrow$) leads to a decrease in financial investment. Since shadow banking activity is increasing in I_f, we conclude that

$$\frac{d \ln l_{f_{i,t}}^{SB}}{d \kappa_{c,t}} < 0,$$

which proves Hypothesis H1.

Appendix A.2. Hypothesis H2

We now establish Hypotheses H2a and H2b, which concern the heterogeneity in the effect of bank competition on shadow banking depending on a firm's investment profile.

In the unconstrained case, the optimal allocation satisfies the first-order condition

$$\phi'(I_r^*) = \theta'(I_f^*; \kappa),$$

with $I_f^* = R - I_r^*$. Differentiating implicitly with respect to κ yields

$$(\phi'' + \theta'') \cdot \frac{\partial I_r^*}{\partial \kappa} = \frac{\partial \theta'}{\partial \kappa}, \quad \frac{\partial I_f^*}{\partial \kappa} = -\frac{\partial I_r^*}{\partial \kappa}.$$

To test H2a, we ask how the effect of κ on I_f^* varies with the level of financial investment. Since firms with higher I_f operate further along the concave θ function, θ'' is more negative. It follows that

$$\frac{\partial^2 I_f^*}{\partial \kappa \, \partial I_f} = \frac{-\partial \theta'' / \partial I_f \cdot \partial I_r^* / \partial \kappa}{\phi'' + \theta''} < 0.$$

This proves H2a: the reduction in financial investment induced by bank competition is stronger among firms that are more financially invested.

For H2b, consider the constrained case where $I_r^* = \lambda(\kappa)$. Then $I_f^* = R - \lambda(\kappa)$, and

$$\frac{\partial I_f^*}{\partial \kappa} = -\lambda'(\kappa) < 0.$$

The strength of this effect depends on how binding the constraint is. Firms with a higher preference for real investment (i.e., steeper ϕ') are more likely to be constrained and more sensitive to a marginal relaxation of the borrowing limit. Thus,

$$\frac{\partial^2 I_r^*}{\partial \kappa \, \partial I_r} > 0, \quad \frac{\partial^2 I_f^*}{\partial \kappa \, \partial I_r} < 0,$$

which confirms H2b: firms with higher real investment intensity reallocate more capital away from financial uses when bank competition increases.

Appendix A.3. Hypothesis H3

We now prove Hypothesis H3, which states that the effect of bank competition on shadow banking differs systematically by ownership type due to asymmetric borrowing constraints and differences in competitive discipline.

Let ownership type be denoted by $o \in \{\text{SOE}, \text{Priv}\}$. Firms solve

$$\max_{I_r, I_f} \quad \phi(I_r) + \theta_o(I_f; \kappa, \rho) \quad \text{s.t.} \quad I_r + I_f = R, \quad I_r \leq \lambda_o(\kappa),$$

where $\theta_o'(I_f; \kappa, \rho) = \theta_0'(I_f) + \psi_o(\rho) - \eta_o(\kappa)$, with $\psi_o' > 0$, $\eta_o' > 0$, and $\theta_0'' < 0$. We assume that SOEs face looser borrowing limits and weaker marginal discipline, i.e.,

$$\lambda_{\text{SOE}}(\kappa) > \lambda_{\text{Priv}}(\kappa), \quad \eta_{\text{SOE}}'(\kappa) < \eta_{\text{Priv}}'(\kappa), \quad \psi_{\text{SOE}}(\rho) > \psi_{\text{Priv}}(\rho).$$

In the unconstrained region, optimality requires

$$\phi'(I_r^*) = \theta_0'(I_f^*) + \psi_o(\rho) - \eta_o(\kappa).$$

Differentiating implicitly with respect to κ gives

$$(\phi'' + \theta_0'') \cdot \frac{\partial I_r^*}{\partial \kappa} = \eta_o'(\kappa), \quad \frac{\partial I_f^*}{\partial \kappa} = -\frac{\partial I_r^*}{\partial \kappa}.$$

Since $\eta'_{\text{Priv}}(\kappa) > \eta'_{\text{SOE}}(\kappa)$, the reduction in financial investment is stronger for private firms:

$$\left|\frac{\partial I_f^*}{\partial \kappa}\right|_{\text{Priv}} > \left|\frac{\partial I_f^*}{\partial \kappa}\right|_{\text{SOE}}.$$

In the constrained region, $I_r^* = \lambda_o(\kappa)$, so $\partial I_f^*/\partial \kappa = -\lambda_o'(\kappa)$. Since $\lambda'_{\text{SOE}}(\kappa) < \lambda'_{\text{Priv}}(\kappa)$, the reallocation is again stronger for private firms.

To evaluate amplification effects, consider the interaction with financial intensity. Since SOEs earn larger arbitrage rents $\psi_o(\rho)$, they allocate more capital to I_f in equilibrium. Due to concavity of θ_0, the marginal decline in θ_0' is smaller at higher I_f, which dampens the reallocation response to κ. Therefore,

$$\frac{\partial^2 I_f^*}{\partial \kappa \partial I_f}\bigg|_{\text{Priv}} > \frac{\partial^2 I_f^*}{\partial \kappa \partial I_f}\bigg|_{\text{SOE}}.$$

Similarly, consider amplification via real intensity. Private firms are more likely to face binding constraints and more responsive to constraint relaxation. Hence,

$$\frac{\partial^2 I_f^*}{\partial \kappa \partial I_r}\bigg|_{\text{Priv}} < \frac{\partial^2 I_f^*}{\partial \kappa \partial I_r}\bigg|_{\text{SOE}},$$

which implies stronger push-channel reallocation for private firms.

Together, these results confirm that both the level effect and the amplification effects of bank competition on shadow banking are stronger for private firms than for SOEs, proving Hypothesis H3.

Appendix B. Tables and figures for Section 4.2

Table B.1
Alternative measures for bank concentration and shadow banking.

	(1) $\ln I_f^{SB}$	(2) $\ln I_f^{SB}$	(3) $\ln I_f^{SB}$	(4) $\ln I_f^{EL}$	(5) $\ln I_f^{EL}$	(6) $\ln I_f^{EL}$
HHI^p	-1.392	-1.456	-2.478			
	(1.425)	(1.429)	(1.521)			
$HHI^p \times \tilde{I}_f$		12.156**				
		(5.190)				
$HHI^p \times \tilde{I}_r$			6.104**			
			(2.650)			
HHI				0.653	0.559	-0.209
				(0.572)	(0.573)	(0.676)
$HHI \times \tilde{I}_f$					6.095	
					(4.383)	
$HHI \times \tilde{I}_r$						4.970**
						(1.949)
$\tilde{I}_f$		-2.663*			-0.164	
		(1.529)			(1.360)	
$\tilde{I}_r$			-1.545*			-1.185*
			(0.805)			(0.653)
Controls	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Industry-Year FE	YES	YES	YES	YES	YES	YES
Observations	24138	24138	24138	24138	24138	24138
Adjusted R^2	0.639	0.639	0.639	0.631	0.632	0.631

Notes: This table presents robustness check results. Columns (1)–(3) use alternative bank competition measure: HHI^p refers to the Herfindahl–Hirschman Index of commercial banks at the provincial level. Columns (4)–(6) employ an alternative shadow banking measure using the natural logarithm of entrusted loans. All other variables remain consistent with those in Table 3. Firm FE represents firm fixed effects, while Industry-Year FE represents industry-by-year fixed effects. Standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1% levels, respectively. Detailed definitions of variables and their data sources can be found in Table 1.

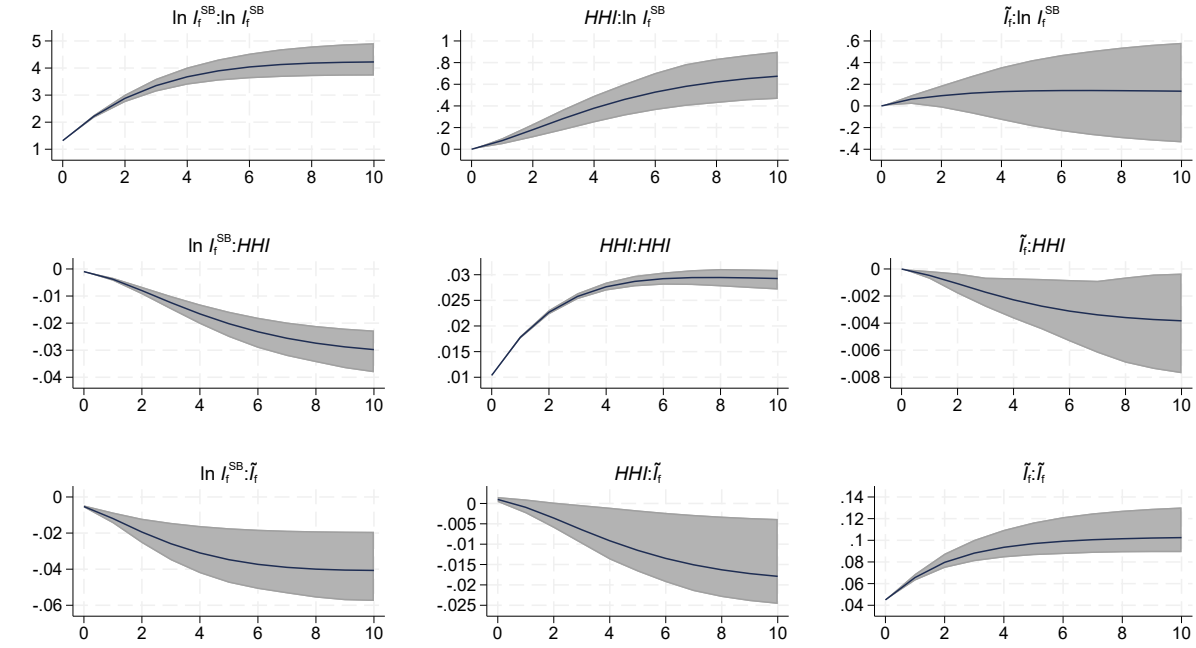


Fig B.1. Impulse response functions (variables: $\ln I_f^{SB}$, HHI, and $\tilde{I}_f$).

Note: Labels in graphs denote impulse:response variable pairs. The estimated system follows the dynamic panel regression $Y_{i,t} = A_1 Y_{i,t-1} + A_2 Y_{i,t-2} + \cdots + A_p Y_{i,t-p} + \mu_i + \lambda_t + \varepsilon_{i,t}$, where $Y_{i,t} = [\ln I_f^{SB}, HHI, \tilde{I}_f]'$ is the vector of endogenous variables, μ_i denotes firm fiexed effects, λ_t year fixed effects, and $\varepsilon_{i,t}$ the residual. The impulse responses confirm the main findings of the paper: bank concentration shocks reduce shadow banking activity over time, particularly when firms are more financially invested, consistent with the disciplining effect of bank competition and diminishing returns to financialisation. Detailed definitions of variables and their data sources can be found in Table 1.

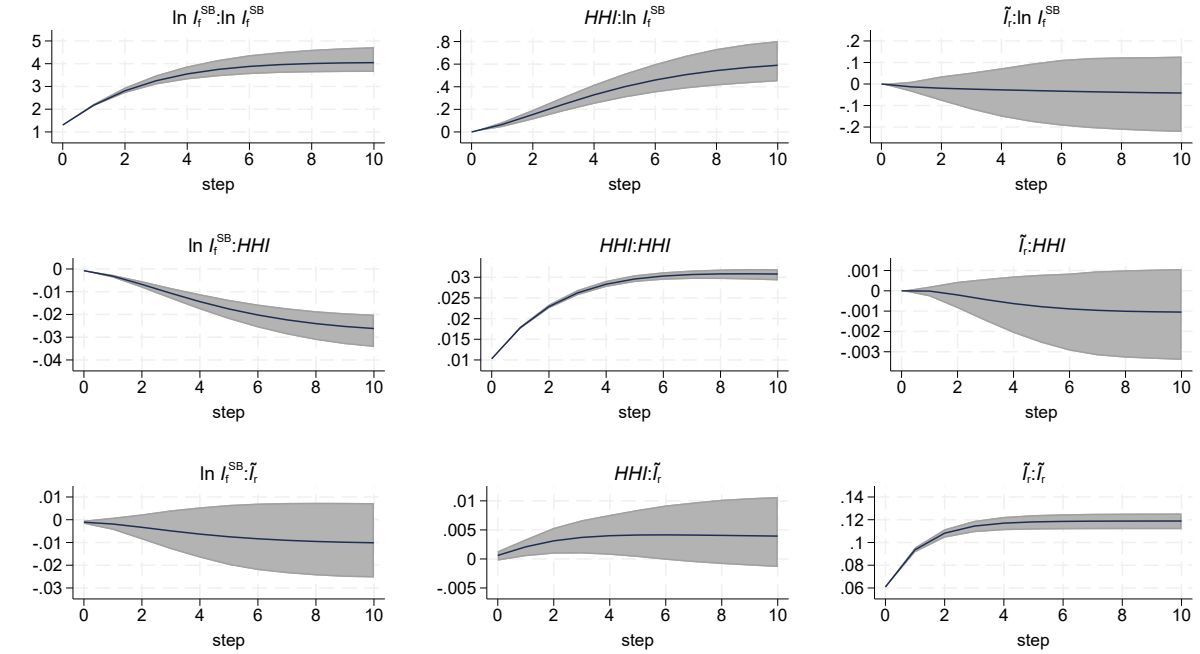


Fig B.2. Impulse response functions (variables: $\ln I_f^{SB}$, HHI, and $\tilde{I}_r$).

Note: Labels in graphs denote impulse:response variable pairs. The estimated system is a panel vector autoregression of the form $Y_{i,t} = A_1 Y_{i,t-1} + A_2 Y_{i,t-2} + \cdots + A_p Y_{i,t-p} + \mu_i + \lambda_t + \varepsilon_{i,t}$, where $Y_{i,t} = [\ln I_f^{SB}, HHI, \tilde{I}_r]'$ includes shadow banking, bank concentration, and real investment ratio. Firm fixed effects μ_i, year fixed effects λ_t, and residuals $\varepsilon_{i,t}$ are included. The impulse responses show that bank concentration shocks lead to a persistent reduction in shadow banking and a delayed increase in real investment, consistent with a pull-based reallocation mechanism. Detailed definitions of variables and their data sources can be found in Table 1.

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